

ABDELRAHMAN IHAB

AI Engineer · Computer Vision · Cybersecurity · Robotics

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PROFESSIONAL SUMMARY

Fourth-year Computer Engineering student at E-JUST specializing in AI, Computer Vision, Cybersecurity, and Robotics. Led computer vision for an award-winning ROV that placed 1st at MATE ROV 2026 Pioneer Level (+ Best Design & Best Presentation). Designed and deployed end-to-end AI systems for malware detection, real-time sports analytics, and biometric detection. Holds 120-hour Fortinet Cybersecurity certification (93%), IBM SkillsBuild Cybersecurity, and Google Cybersecurity credentials. Proven track record across 5 industry internships in network administration, cybersecurity, and IT operations. Seeking internship or research roles in AI engineering, computer vision, or offensive/defensive cybersecurity.

EDUCATION

B.Sc. Computer Engineering | Egypt-Japan University of Science and Technology (E-JUST) *Oct 2022 – Present (Expected 2027)*

- AI & Intelligent Systems: Machine Learning, Deep Learning, Computer Vision, Pattern Recognition
- Security & Systems: Embedded Systems, Operating Systems, Parallel & Distributed Computing
- Core Engineering: Algorithms, Data Structures, Theory of Computation, Computer Networks, Robotics

IGCSE — Grade: 99.5% | Alexandria International School *Oct 2008 – Jun 2022*

EXPERIENCE

Computer Vision / ML Engineer | Vortex Robotics (Seasonal — Competition Team) *Jan 2026 – Present*

- Designed and deployed a full YOLOv8 computer vision pipeline for underwater ROV competition; achieved **>95% trajectory continuity**
- Built European Green Crab detection system using synthetic data augmentation to overcome a small real-world dataset, enabling reliable biodiversity analysis in challenging underwater conditions.
- Developed a 3D coral garden reconstruction model with real-world spatial calibration and accurate metric scaling for scientific reporting.
- Team secured **1st Place, Best Design Award, and Best Presentation Award**
- at MATE ROV 2026 Pioneer Level Regional Competition and qualified for the international competition in Canada.

Fortinet Cybersecurity Trainee | National Telecom Institute (NTI) · ITIDA *Aug–Sep 2025 · 120 hrs*

- Completed 120-hour intensive dual-track program (30 hrs soft skills + 90 hrs technical), achieving **93% overall score**.
- Configured FortiGate firewalls: security policies, NAT, FSSO authentication, and routing protocols from scratch.
- Implemented and tested SSL VPN, IPSec VPN, SD-WAN, Security Fabric, IPS, antivirus, web filtering, and High Availability setups.
- Studied cryptography, PKI, certificate lifecycle management, and intrusion prevention in production-simulated environments.

IT Internship Trainee | Ezz Dekheila Steel Company (EZDK)

Jul–Aug 2025

- Gained enterprise IT operations exposure within one of Egypt's largest industrial organizations.
- Applied networking, cybersecurity, and web development skills to real production infrastructure and IT workflows.
- Observed and assisted with corporate network management, cloud computing practices, and project coordination.

Computer Engineer Intern | PETROJET

Aug 2025

- Diagnosed and resolved network device failures across enterprise infrastructure, maintaining system uptime and reliability.
- Gained hands-on experience with SAP enterprise systems; collaborated with cross-functional IT teams on operational tasks.
- Documented network configurations and troubleshooting procedures to support knowledge transfer within the team.

Network Administrator Intern | DIECO — Dhanab Integrated Engineering Company

Jul–Sep 2024

- Assisted in the full lifecycle of network infrastructure setup: planning, configuration, installation, and ongoing maintenance.
- Configured and maintained routers, switches, and VPN endpoints; monitored performance metrics to ensure SLA compliance.
- Supported installation of Wide Area Network (WAN) and local network devices across multiple office locations.

Cybersecurity Intern | Egyptian Ethylene & Derivatives Company (ETHYDCO)

Jul 2024

- Identified and documented security vulnerabilities across industrial control systems; proposed and implemented mitigation measures including firewall rule updates and access control policies.
- Conducted vulnerability scans using standard tools; contributed to incident monitoring, threat detection, and response workflows.
- Assisted in enforcing authentication controls and access management policies aligned with cybersecurity best practices.

PROJECTS

Nautix — Underwater ROV Computer Vision System [1st Place · Best Design · Best Presentation — MATE ROV 2026]

github.com/Abdelrahman-288/Mate-ROV-Pioneer-2026

- Engineered a YOLOv8 object detection pipeline for European Green Crab identification; applied synthetic data augmentation and custom training to maximize accuracy on a severely limited real-world dataset.
- Implemented linear interpolation for occluded frame recovery, achieving **>95% trajectory continuity** across all test dives.
- Built a 3D coral garden reconstruction system using photogrammetry-based techniques with calibrated real-world metric scaling, enabling scientifically usable biodiversity output.
- **Stack:** Python · YOLOv8 · OpenCV · NumPy · Synthetic Data Generation · Homography · 3D Reconstruction

Hybrid AI Malware Classification & Detection System [Deep Learning · Cybersecurity · Static & Dynamic Analysis]

github.com/Abdelrahman-288/malware-classification-system

- Architected a 4-stage hybrid pipeline on the Microsoft BIG 2015 dataset (9 malware families, 10,868 EXE samples): binary visualization → CNN feature extraction → opcode sequence → BiLSTM → ensemble fusion.
- Performed static malware analysis using pefile for PE header inspection and Ghidra for opcode sequence extraction and disassembly; enriched feature set with entropy analysis and behavioral metadata.
- Integrated Grad-CAM explainability layer to provide visual justification for predictions, increasing model interpretability for security analysts.
- Deployed an interactive Streamlit dashboard for real-time file scanning, prediction output, visualization, and threat reporting with automated feature engineering pipeline.
- **Stack:** Python · TensorFlow · CNN · BiLSTM · pefile · Ghidra · Streamlit · Grad-CAM · NumPy · Pandas

Tennis Match Analytics — End-to-End Computer Vision Pipeline [Real-Time Sports Analytics · YOLOv8 · OpenCV]

github.com/Abdelrahman-288/Tennis-Analysis

- Fine-tuned YOLOv8 for player and ball detection on single fixed-camera video, improving ball detection AP by **18%**
- over baseline.
- Applied homography-based court plane mapping and trajectory interpolation for rally metric computation, achieving **93.31% accuracy**
- on derived statistics.
- Built a custom CNN with keypoint regression for robust court boundary detection, eliminating dependency on manual calibration.
- System outputs shot direction, player speed, displacement, and rally results as annotated video overlays — a cost-effective alternative to professional tracking systems.
- **Stack:** Python · YOLOv8 · PyTorch · OpenCV · Homography · CNN · Keypoint Regression

Deep Face Detection System — VGG16 + Gradio [Transfer Learning · Object Detection · Web Deployment]

github.com/Abdelrahman-288/Deep_Face_Detection_Project

- Built a custom face detection model by fine-tuning a pretrained VGG16 backbone with additional bounding-box regression and binary classification heads for face localization.
- Generated and annotated a training dataset from scratch using OpenCV-based capture and LabelMe for polygon/bounding-box annotation; exported in VOC and COCO formats.
- Deployed the trained model as an interactive Gradio web application supporting real-time inference from uploaded images or live webcam feed.
- **Stack:** Python · TensorFlow · VGG16 · OpenCV · LabelMe · Gradio

Heart Disease Prediction — End-to-End ML Pipeline [Supervised Learning · Feature Engineering · Model Selection]

github.com/Abdelrahman-288/Heart_Disease_Prediction

- Built a complete ML pipeline on the UCI Heart Disease dataset: data cleaning, exploratory analysis, PCA-based dimensionality reduction, and feature selection.
- Trained and compared four models (Logistic Regression, Decision Trees, Random Forest, SVM); optimized hyperparameters using GridSearchCV and RandomizedSearchCV.
- Applied unsupervised clustering to surface patient subgroups, supplementing supervised predictions with population-level insight.
- **Stack:** Python · Scikit-learn · Pandas · NumPy · Matplotlib · Seaborn · Streamlit

Dual-Frequency RFID Defense Architecture [Graduation Project · Hardware Security · Protocol Design]

- Designed a physical anti-cloning RFID security system combining dual HF (13.56 MHz) + UHF (860–960 MHz) chips on a single card with a dual-antenna reader and central collision-detection server.
- Implemented rolling session tokens, dynamic time-delta (Δt) fraud detection, and a distance bounding protocol to prevent relay and replay attacks.
- Built graceful degradation fallback so the system remains operational when one frequency band is unavailable; demonstrated full attack and defense scenarios using Proxmark3.
- **Stack:** Proxmark3 · HF/UHF RFID · Embedded C · Security Protocol Design · Anti-Relay / Anti-Clone Architecture

PROFESSIONAL TRAINING & PROGRAMS

Fortinet Cybersecurity Program | NTI · ITIDA

Aug–Sep 2025 · 120 hrs · Score: 93%

- Network fundamentals: IP addressing, subnetting, device configuration, routing protocols.
- FortiGate configuration: firewall policies, NAT, FSSO authentication, FortiManager integration.
- VPN implementation (SSL & IPSec), SD-WAN, Security Fabric, High Availability and diagnostics.
- Threat protection: IPS, antivirus, web filtering, application control, cryptography, and PKI.

NVIDIA AI Program | ITI

Sep–Oct 2025

- Completed: Building LLM Applications with Prompt Engineering — prompt design, LLM optimization, API integration.
- Completed: AI for All: From Basics to Generative AI — AI fundamentals, model training, GenAI applications.

Embedded Linux Training Program | IEEE

Sep–Oct 2025

- Built strong Linux foundations: file systems, permissions, logs, networking tools, and Bash scripting.
- Developed device drivers (GPIO, IOCTL, sysfs) on Raspberry Pi using C and Python.
- Embedded Linux essentials: Yocto Project, Docker containerization, debugging with GDB and Valgrind.

AI & Machine Learning Program | Sprints

Aug–Oct 2025 · 40+ hrs

- Mastered Python (OOP, data structures, error handling) and advanced data analysis with NumPy, Pandas, Matplotlib, and Seaborn.
- Built a full supervised + unsupervised ML pipeline for Heart Disease Prediction (Logistic Regression, Random Forest, SVM, PCA, GridSearchCV).

IBM SkillsBuild Cybersecurity Certificate | IBM

Jun–Aug 2025 · 60+ hrs

- Governance, Risk & Compliance (GRC), Data Privacy, and Vulnerability Management.
- SOC operations, Incident Response, Digital Forensics, Cloud Security architecture and capstone project.

TECHNICAL SKILLS

Programming	Python (primary) · C++ · C · Bash · HTML/CSS · JavaScript
AI & Deep Learning	TensorFlow · PyTorch · Scikit-learn · CNN · BiLSTM · VGG16 · Grad-CAM · Transfer Learning
Computer Vision	YOLOv8 · OpenCV · Homography Mapping · 3D Reconstruction · LabelMe · Gradio · Synthetic Data
Cybersecurity	FortiGate · SD-WAN · SSL/IPSec VPN · IPS/IDS · Malware Analysis · Ghidra · pefile · Proxmark3 · Wireshark
Data & Analytics	NumPy · Pandas · Matplotlib · Seaborn · Plotly · Streamlit · Jupyter Notebook
Systems & Linux	Embedded Linux · Yocto · Raspberry Pi · Device Drivers · Docker · GDB · Valgrind · Bash Scripting
Networking	TCP/IP · Routing Protocols · Firewall Configuration · VPN · WAN Setup · Network Troubleshooting
Cloud & DevOps	Git · GitHub · AWS (fundamentals) · Google Cloud (fundamentals)
Robotics & HW	ROV Vision Systems · RFID Security (HF/UHF) · Sensor Integration · Proxmark3

CERTIFICATIONS & CREDENTIALS

MATE ROV 2026 — Pioneer Level Regional Champion — MATE ROV Competition	Apr 2026
Universal Robots UR20/30 eLearning — Universal Robots Academy	Apr 2026
Play It Safe: Manage Security Risks — Google · Coursera	May 2026
Foundations of Cybersecurity — Google · Coursera	Mar 2026
Hands-on Introduction to Linux Commands & Shell Scripting — IBM · Coursera	Feb 2026
Fortinet Cybersecurity Training — Score: 93% — NTI · ITIDA	Sep 2025
IBM SkillsBuild Cybersecurity Certificate — IBM SkillsBuild	Aug 2025
Building LLM Applications with Prompt Engineering — NVIDIA	Jul 2025
AI for All: From Basics to Generative AI — NVIDIA	Jul 2025
AI & Machine Learning Foundations — Sprints	Aug 2025
Embedded Linux Course — IEEE	Dec 2025
Stanford CS229 — Machine Learning — Coursera · Stanford University	Completed
English Language Certificate — Oxford	2022